Many motions we see in our day to day life are non-uniform. We will learn more about uniform and non-uniform motion in higher classes.

If an object covers uniform distances in uniform intervals then the motion of the object is called uniform motion. Otherwise the motion is called non-uniform motion.

In a nutshell, we can classify the motion in terms of a) path b) if it is periodic or not c) if the speed is uniform or not. However, in real life, the motions are combinations of many types of motion.

Multiple Motion

Look at the bicycle in the picture. What type of motion does the wheel perform? What type of motion does the cycle in total perform?

The tyres rotate and make a rotatory motion, but the cycle as such moves forward in a linear path.

Activity 7

Simple Spinner

Let us enjoy by making a simple spinner. Make it by the following instruction.

Cut a 2cm long piece from an old ball-pen refill and make a hole in its center with a divider point (Fig. 1).

Take a thin wire of length 9cm and fold it into a U-shape (Fig. 2).

Weave the refill spinner in the U-shaped wire (Fig. 3).

Wrap the two ends of the wire on the plastic refill, leaving enough clearance for the spinner to rotate (Fig. 4).

On blowing through the refill, the spinner rotates (Fig. 5).

For obtaining maximum speed adjust the wires so that air is directed towards the ends of the spinner.

Have you enjoyed with simple spinner? Do you observe the motions in the toy? Can you answer the following questions?

1. Motion of the air in tube is _____ motion.
2. Motion of the refill stick is _____ motion.
3. The toy converts _____ motion into _____ motion.

Think

In a simple spinner linear motion is converted into rotatory motion. Can you make a toy which converts rotatory motion into linear motion?